MIL-B-50451B(PA)
1 JULY 1976
SUPERSEDING
MIL-B-50451A(MU)
15 OCTOBER 1971

MILITARY SPECIFICATION

BLASTING MACHINE, M32
PARTS AND ASSEMBLY

This specification is approved for use by Picatinny Arsenal, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the parts and assembly for one type of blasting machine designated as Blasting Machine, M32.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing
PPP-B-621	- Boxes, Wood, Nailed and Lock-corner
PPP-B-636	- Box, Fiberboard

MILITARY

MIL-P-116	- Preservation, Methods of
MIL-B-117	- Bag, Sleeve and Tubing - Interior Packaging
MIL-S-4473	- Shielding Magnetron Tubes and Magnets for Air Shipment
MIL-B-46506-	Box, Ammunition Packing Wood, Wire-bound
*MIL-A-48078-	Ammunition, Standard Quality Assurance Provisions, General Specification For
MIL-F-50449-	Filler Sheet Form (For use in Ammunition Containers)

FSC: 1375

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STANDARDS

FEDERAL

FED-STD-151- Metals, Test Methods

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-129 - Marking For Shipment and Storage
MIL-STD-1169- Packaging, Packing and Marking For Shipment of Inert Ammunition Components

*For meaning of asterisk (*) see paragraph 6.4.

DRAWINGS

U.S. ARMY PICATINNY ARSENAL

9233869 - M32 Blasting Machine
9235504 - Blaster Assembly
9235509 - Internal Assembly
9235513 - Alternator Assembly
9235518 - Gear, Pinion Rotor
9235519 - Stator Assembly
9235523 - Network, Pulse Delay
9235529 - Gear Subassembly
9235530 - Gear, Pinion
9235531 - Gear, Spur
9235532 - Mechanical Housing Assembly
9235534 - Gear Assembly
9235535 - Gear Rack Assembly
9235537 - Gear, Pinion
9235546 - Gear, Spur
9235547 - Barrel, Clutch
9235549 - Clutch

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting officer.)

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3. REQUIREMENTS

3.1 Materials and parts. - Materials and parts shall be in accordance with applicable drawings and specifications.

3.2 Assembly. - The assembly shall comply with all requirements specified on Drawing (Dwg. 9233869) and with all requirements specified in applicable specifications.

3.3 Performance

3.3.1 Capacity. - The blasting machine shall be capable of producing a minimum output of 1.5 amperes through an external resistance of 48 ohms plus 10 percent for a duration of not less than 1.5 milliseconds (See 4.5.1).

3.3.2 Life cycle. - The blasting machine shall be capable of producing the output specified in 3.3.1 for not less than 5000 successive times without any component replacement or repair (See 4.5.2).

3.3.3 Temperature and humidity conditions. -

3.3.3.1 Operating temperature. - The blasting machine shall withstand operating temperatures of minus 40 degrees Fahrenheit (F) and plus 125 degrees (F) without damage, and shall operate within the requirements of 3.3.1 (See 4.5.10).

3.3.3.2 Humidity test. - The blasting machine shall operate within the requirements of 3.3.1 after exposure to 20 hours of relative humidity between 92 and 97 percent at air temperatures from plus 80 degrees F to plus 85 degrees F, followed by 4 hours of 100 percent relative humidity with condensation at plus 75 degrees F to plus 80 degrees F (See 4.5.3).

3.3.4 Four-foot drop. - The blasting machine shall withstand being dropped a vertical distance of 4 feet through each of its three major axes on a solid concrete surface (See 4.5.5).

3.3.5 Insulation resistance. - The blasting machine shall have greater than 30 megohms of resistance at 500 volts d.c. when measured from the shorted output terminals to the plunger (See 4.5.6).

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*3.3.6 Leakage and Operability. - The blasting machine shall show no evidence of leakage and shall be operable after immersion in water (See 4.5.7).

3.4 First article inspection. - This specification contains technical provisions for first article inspection. Requirements for the submission of first article samples by the contractor shall be as specified in the contract.

3.5 Workmanship. - All parts and assemblies shall be fabricated and finished in a thorough workmanlike manner and all manufacturing, processing, and assembly operations shall be correctly performed. They shall be free of burrs, chips, sharp edges, cracks, crazes, unblended radii, porosity, warpage, burn marks, checks, chipped edges, blisters, excess flash, dirt, grease, rust, salt deposits, visible raised cement seams, solder splash, corrosion products, and other defects and foreign matter which would affect their serviceability. The cleaning method used shall not be injurious to any part or assembly nor shall the parts be contaminated by the cleaning agent. All required markings shall be neat, legible and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

*4.1 Responsibility for inspection and standard quality assurance provisions. - Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspections. - The following types of inspection shall be conducted on this item.

a. First Article Inspection
b. Quality Conformance Inspection

4.3 First article inspection

*4.3.1 Submission. - The contractor shall submit a first article sample (See 6.1b) as designated by the Contracting Officer. The first article sample shall consist of eighteen (18) complete assemblies and fifteen (15) complete sets of parts (i.e. 15 each of every component and sub-assembly, including at least five (5) items from each cavity used to produce molded parts, and be so identified as to permit tracing

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each item to the cavity in which it was molded). The first article sample shall be evaluated in accordance with the provisions of 4.3.2 and 4.3.3.

*4.3.2 Inspections to be performed. - See MIL-A-48078, 4.4.2, testing per 4.4.3, molding per 4.4.5 and Table I specified herein.

*4.3.3 Rejection. - (See MIL-A-48078).

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
*	Blasting Machine, M32				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Parts - Examination for defects	(a)			
	Blaster Assembly (see dwg 9235504)	15 ea		3.1	4.4.2.1 through 4.4.2.3
	Internal Assembly (see dwg 9235509)	15 ea		3.1	4.4.2.4 through 4.4.2.7
	Alternator Assembly (see dwg 9235513)	15 ea		3.1	4.4.2.8 through 4.4.2.13
	Stator Assembly (see dwg 9235519)	15 ea		3.1	4.4.2.14
	Network Pulse delay (see dwg 9235523)	15 ea		3.1	4.4.2.15
	Gear Sub-Assembly (see dwg 9235529)	15 ea		3.1	4.4.2.16 and 4.4.2.17
	Mechanical Housing Assembly (see dwg 9235532)	15 ea		3.1	4.4.2.18 through 4.4.2.23
	Blaster Assembly (see dwg 9233869)	15 ea		3.1	4.4.2.24
	Gear Assembly (see dwg 9235534)	15 ea		3.1	4.4.2.25 and 4.4.2.26
	Testing - Component Parts				
Major	Resistance test of stator assembly (see dwg 9235519)	15			4.5.8
Major	Clutch torque test (see dwg 9235549)	15			4.5.9
Major	Hardness of spur gear (see dwg 9235546)	15			4.5.4
Major	Hardness of spur gear (see dwg 9235531)	15			4.5.4
Major	Hardness of pinion gear (see dwg 9235518)	15			4.5.4
Major	Hardness of pinion gear (see dwg 9235530)	15			4.5.4
Major	Hardness of clutch barrel (see dwg 9235547)	15			4.5.4
Major	Hardness of gear rack assembly (see dwg 9235535)	15			4.5.4
Major	Hardness of pinion gear (see dwg 9235537)	15			4.5.4

NOTES:

(a) Individual parts enumerated on sub-assembly drawings.

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TABLE I - FIRST ARTICLE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
*	Blasting Machine, M32				See Below
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
	Testing - Blasting Machine M32				
	Four-foot drop	3		3.3.4	4.5.5
	Humidity test	3		3.3.3.2	4.5.3
	Operating Temperature	3		3.3.3.1	4.5.10
	Life cycle	3		3.3.2	4.5.2
	Insulation resistance	3		3.3.5	4.5.6
	Leakage and Operability	3		3.3.6	4.5.7
	Molded Parts				
	Housing Electrical (see dwg 9235541)	5(b)		3.1	4.4.5.1
	Housing Mechanical drive (see dwg 9235533)	5(b)		3.1	4.4.5.2
	Case, Molding (see dwg 9235505)	5(b)		3.1	4.4.5.3
	Lever (see 9235508)	5(b)		3.1	4.4.5.4

NOTES:

(b) 5 per cavity (see 4.3.1)

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4.4 Quality conformance inspection. -

*4.4.1 Inspection lot formation. - Inspection lots shall comply with the lot formation provisions of MIL-A-48078. In addition, inspection lots of blasting machines shall contain parts from not more than one lot interfix number from one manufacturer.

*4.4.2 Examination. - (See MIL-A-48078). Unless otherwise specified in the classification of defects and test tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.4.4.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235505
*4.4.2.1	Case, Molding				NEXT HIGHER ASSEMBLY 9235504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crack or split		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235508
*4.4.2.2	Lever				NEXT HIGHER ASSEMBLY 9235504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crack or split		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235509
*4.4.2.3	Internal Assembly				NEXT HIGHER ASSEMBLY 9235504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Soldering missing or inadequate		0.40%	3.1	Visual
102	Application of adhesive improper		0.40%	3.1	Visual
103	Coating missing or inadequate		0.40%	3.1	Visual
104	Capacitor exposed above surface of epoxy		0.40%	3.1	Visual
105	Capacitor leads improperly attached to proper standoff terminal		0.40%	3.1	Visual - Manual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235514
*4.4.2.4	Plate, Bearing				NEXT HIGHER ASSEMBLY 9235509
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of small outside diameter with bearing diameter		0.40%	3.1	Gage
102	Small outside diameter		0.40%	3.1	Gage
103	Bearing diameter		0.40%	3.1	Gage
Minor					
201	Chamfer missing		0.65%	3.1	Visual
202	Overall thickness		0.65%	3.1	Gage
203	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235541
*4.4.2.5	Housing Electrical				NEXT HIGHER ASSEMBLY 9235509
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Crack or split		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.6	Mechanical Housing Assembly				9235532
					NEXT HIGHER ASSEMBLY
					9235509
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Component missing		0.40%	3.1	Visual
102	Lubrication missing		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET	OF	DRAWING NUMBER
*4.4.2.7	Network, Pulse Delay		1	1	9235523
					NEXT HIGHER ASSEMBLY 9235509
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Parallelism of flat standoff with center line		0.40%	3.1	Gage
102	Component missing or improperly located		0.40%	3.1	Visual
103	Wiring improper		0.40%	3.1	Visual
104	Soldering missing or inadequate		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235517
*4.4.2.8	End Bell, Drive				NEXT HIGHER ASSEMBLY 9235513
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of rotor sleeve diameter with small inside diameter		0.40%	3.1	Gage
102	True position of bearing cavity diameter with small inside diameter		0.40%	3.1	Gage
103	True position of small outside diameter with small inside diameter		0.40%	3.1	Gage
104	Rotor sleeve diameter		0.40%	3.1	Gage
105	Diameter of bearing cavity		0.40%	3.1	Gage
106	Small inside diameter		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
*4.4.2.9	Sleeve, Rotor	1	1	9235516
				NEXT HIGHER ASSEMBLY 9235513

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of inside diameter with outside diameter		0.40%	3.1	Gage
102	Inside diameter		0.40%	3.1	Gage
Minor					
201	Outside diameter		0.65%	3.1	Gage
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235520
*4.4.2.10	End Bell				NEXT HIGHER ASSEMBLY 9235513
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of outside diameter with bearing diameter		0.40%	3.1	Gage
102	True position of large inside diameter with bearing diameter		0.40%	3.1	Gage
103	Large inside diameter		0.40%	3.1	Gage
104	Bearing diameter		0.40%	3.1	Gage
Minor					
201	Length		0.65%	3.1	Gage
202	Thickness of bottom		0.65%	3.1	Gage
203	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235518
*4.4.2.11	Gear, Pinion Rotor				NEXT HIGHER ASSEMBLY 9235513
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Outside diameter		0.40%	3.1	Gage
102	Face width of pinion, max.		0.40%	3.1	Gage
103	Total length		0.40%	3.1	Gage
104	Total composite error of gear		0.40%	3.1	Gage
105	Tooth to tooth composite error		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.12	Magnet, Permanent				9235515
					NEXT HIGHER ASSEMBLY 9235513
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of inner diameter to outside diameter		0.40%	3.1	Gage
Minor					
201	Inside diameter		0.65%	3.1	Gage
202	Outside diameter		0.65%	3.1	Gage
203	Evidence of poor workmanship		1.00%	3.5	Visual

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235519
*4.4.2.13	Stator Assembly				NEXT HIGHER ASSEMBLY 9235513
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of outside diameter of stator assembly to shoulder of shaft		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.14	Shaft, Shouldered				9235542
					NEXT HIGHER ASSEMBLY
					9235519
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of small outside diameter with intermediate outside diameter		0.40%	3.1	Gage
102	Intermediate outside diameter		0.40%	3.1	Gage
103	Inside diameter		0.40%	3.1	Gage
104	Small outside diameter		0.40%	3.1	Gage
105	Outside diameter of spline		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.15	Standoff				9235528
					NEXT HIGHER ASSEMBLY 9235523
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Width across flats		0.40%	3.1	Gage
102	Small outside diameter		0.40%	3.1	Gage
103	Small inside diameter		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.16	Gear, Pinion		9235530
			NEXT HIGHER ASSEMBLY 9235529

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of small outside diameter with large outside diameter		0.40%	3.1	Gage
102	Total composite error of gear		0.40%	3.1	Gage
103	Large outside diameter		0.40%	3.1	Gage
104	Small outside diameter		0.40%	3.1	Gage
105	Tooth to tooth composite error		0.40%	3.1	Gage
Minor					
201	Total length		0.65%	3.1	Gage
202	Face width of pinion		0.65%	3.1	Gage
203	Length from small end to end of pinion		0.65%	3.1	Gage
204	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.17	Gear, Spur				9235531
					NEXT HIGHER ASSEMBLY 9235529
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.1	Gage
102	Total composite error of gear		0.40%	3.1	Gage
103	Tooth to tooth composite error		0.40%	3.1	Gage
104	Thickness		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.18	Gear Rack Assembly				9235535
					NEXT HIGHER ASSEMBLY
					9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of outside diameter of plunger with outside diameter of gear rack		0.40%	3.1	Gage
102	Outside diameter of plunger		0.40%	3.1	Gage
103	Outside diameter of gear rack		0.40%	3.1	Gage
104	Diameter over gage pin		0.40%	3.1	Gage
Minor					
201	Number of teeth		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.19	Spring, Helical Compression		9235539
			NEXT HIGHER ASSEMBLY 9235532

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Load at 1.788 deflection		0.40%	3.1	Test
102	Outside diameter		0.40%	3.1	Gage
103	End not closed, ground or squared		0.40%	3.1	Visual
104	Improper number of coils		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.20	Housing, Mechanical Drive				9235533
					NEXT HIGHER ASSEMBLY 9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Pitch diameter of threads, max		0.40%	3.1	Gage
102	Minor diameter of threads, max		0.40%	3.1	Gage
103	Crack or split		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.21	Gear, Pinion		9235537
			NEXT HIGHER ASSEMBLY
			9235532

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of small outside diameter with large outside diameter		0.40%	3.1	Gage
102	Total composite error of gear		0.40%	3.1	Gage
103	Small outside diameter		0.40%	3.1	Gage
104	Large outside diameter		0.40%	3.1	Gage
105	Tooth to tooth composite error		0.40%	3.1	Gage
Minor					
201	Total length		0.65%	3.1	Gage
202	Face width of pinion, max		0.65%	3.1	Gage
203	Length from large end to end of pinion		0.65%	3.1	Gage
204	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235534
*4.4.2.22	Gear Assembly				NEXT HIGHER ASSEMBLY 9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Component missing or improperly assembled		0.40%	3.1	Visual-Manual
102	Stake missing or inadequate		0.40%	3.1	Visual-Manual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.23	Gear Sub-Assembly				9235529
					NEXT HIGHER ASSEMBLY 9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Perpendicularity of gear to shaft		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235504
*4.4.2.24	Blaster Assembly				NEXT HIGHER ASSEMBLY 9233869
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Component missing		0.40%	3.1	Visual
102	Spring in binder post functions improperly		0.40%	3.1	Visual
103	Lever assembled improperly to gear rack		0.40%	3.1	Visual-Manual
104	Lever clip does not restrain handle when handle is in closed position		0.40%	3.1	Visual
105	Retaining ring missing or improperly installed		0.40%	3.1	Visual-Manual
Minor					
201	Identification plate illegible		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9235547
*4.4.2.25	Barrel, Clutch				NEXT HIGHER ASSEMBLY 9235534
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	True position of large inside diameter with small outside diameter		0.40%	3.1	Gage
102	Large inside diameter		0.40%	3.1	Gage
103	Large outside diameter		0.40%	3.1	Gage
104	Small outside diameter		0.40%	3.1	Gage
Minor					
201	Total length		0.65%	3.1	Gage
202	Depth of cavity		0.65%	3.1	Gage
203	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.26	Gear, Spur				9235546
					NEXT HIGHER ASSEMBLY 9235534
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Inside diameter		0.40%	3.1	Gage
102	Total composite error of gear		0.40%	3.1	Gage
103	Tooth to tooth composite error		0.40%	3.1	Gage
104	Thickness		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.27	Box, Fiberboard Prior to Closing				See 5.1.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Number of blasting machines incorrect		0.40%	3.1	Visual
102	Box damaged		0.40%	3.1	Visual
Minor					
201	Packing material missing		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.28	Box, Fiberboard		See 5.1.1
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Box improperly taped		0.40%	3.1	Visual
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual
202	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER
*4.4.2.29	Sealed Packed Barrier Bag, Prior to Packing in Box				See 5.1.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Waterproofness of barrier bag destroyed by rupture, puncture or separation of heat seal		0.40%	3.1	Visual
102	Evidence of crushed box		0.40%	3.1	Visual
103	All excess air not exhausted from bag prior to sealing		0.40%	3.1	Visual
NOTE	Package should be opened if there is evidence crushing				
Minor					
201	Heat seal of barrier bag located so that one or more subsequent seals cannot be applied		0.65%	3.1	Visual
202	Marking of bag misleading or unidentifiable		0.65%	3.1	Visual
203	Evidence of poor workmanship		1.00%	3.5	Visual

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.30	Box, Wood Prior to Closing				See 5.2.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Fiberboard box missing		0.40%	3.1	Visual
102	Board broken or split		0.40%	3.1	Visual
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B(PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.2.31	Box, Wood				See 5.2.1
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Strapping missing, broken or loose		0.40%	3.1	Visual
102	Magnetic caution label missing (when applicable)		0.40%	3.1	Visual (See 5.3)
Minor					
201	Marking missing, misleading or unidentifiable		0.65%	3.1	Visual
202	Strapping improperly assembled		0.65%	3.1	Visual
203	Contents move when box is tilted		0.65%	3.1	Manual
204	Evidence of poor workmanship		1.00%	3.5	Visual

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4.4.3 Testing

4.4.3.1 Component parts. - The following tests for the individual components will be performed as specified below using test equipment specified in 4.4.4. Failure of one or more components to meet the requirements shall reject the lot.

Test Description	Test Method	Test Quantity	Category
Resistance test of stator assembly (See Dwg. 9235519)	4.5.8	20	Major
Clutch torque test (See Dwg. 92355469	4.5.9	20	Major
Hardness of spur gear (See Dwg. 9235546)	4.5.4	20	Major
Hardness of spur gear (See Dwg. 9235531)	4.5.4	20	Major
Hardness of pinion gear (See Dwg. 9235518)	4.5.4	20	Major
Hardness of pinion gear (See Dwg. 9235530)	4.5.4	20	Major
Hardness of clutch barrel (See Dwg. 9235547)	4.5.4	20	Major
Hardness of gear rack assembly (See Dwg. 9235535)	4.5.4	20	Major
Hardness of pinion gear (See Dwg. 9235537)	4.5.4	20	Major

4.4.3.2 Assemblies. - To be tested in sequence indicated below.

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*4.4.3.2.1 Leakage and operability test (See 3.3.6). - Major defect - This test shall be performed 100 percent. Any blasting machine which fails to comply with the requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.7.

*4.4.3.2.2 Capacity (See 3.3.1) - Major defect - This test shall be performed 100 percent. Any blasting machine which fails to comply with the requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.5.1 except that a random sample of twenty (20) from each lot shall require a photograph of the oscilloscope display.

4.4.3 Heat seal test of sealed bag (See 5.1.1). - Major defect. - Sampling, acceptance criteria and test procedures for this test shall be as specified in MIL-P-116.

*4.4.4 Inspection equipment. - The inspection equipment required to perform the inspections and tests prescribed in this specification is identified in the 'Paragraph Reference/ Inspection Method' column in the tables starting with paragraph 4.4.2.1, and the test method paragraphs (See 4.5). The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078, and 6.2 herein.

*4.4.5 Dimensional control of molded parts. - (The requirements of this paragraph apply to those dimensions which are considered "as molded", and which do not undergo subsequent operations). In place of normal sampling and inspections associated with the Classification of Defects, and with the approval of the Contracting Officer, and after a curing time for the parts has been approved (See Note 1) a sample of five (5) parts (as molded) from each cavity shall be inspected dimensionally to qualify a new or reworked cavity for use in production. The molded parts shall carry the individual cavity identification, when more than one cavity is used in production. As a control of each cavity during production, the above quantity of parts from each cavity shall be inspected for a least the defects listed in paragraphs 4.4.5.1 through 4.4.5.4 after continuous production of each 5,000 parts or at the end of each week, whichever occurs first. Of the five (5) samples, one (1) sample shall be the last part produced. If any defective parts are found during qualification of the cavity,

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the cavity producing the defective part will not be used in production. If any defective parts are found when inspection is performed for the control of the cavity, the cavity producing the defective part shall be removed from production. Further, that portion of production since the last control check shall be returned to the contractor for inspection for each separate type of defect according to MIL-STD-105, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. All cavities removed from production because of some fault, may after reworking, be returned to production providing they pass the qualification test above. The contractor may request a change of inspection frequency providing he presents objective evidence to the contracting officer to substantiate the request. Contractor design of gages and test equipment required to perform the inspections listed herein shall be forwarded to Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T-T for approval prior to manufacture of equipment. The noted subparagraphs identify those items and minimum inspections subject to the requirements of this paragraph.

NOTE 1 - In establishing the curing time, dimensionally check ten (10) parts from each cavity at periodic intervals (e.g. every 30, 60, etc. mins.) until dimensional stability is attained. The curing time will be from the time the part comes out of the mold until dimensional stability is attained. The inspection data used in determining the curing time shall be sent to Picatinny Arsenal, Dover, New Jersey 07801 ATTN: SARPA-QA-A-M. If any new change in material, cycle time, a reworked cavity, or a new cavity is used, a new curing time shall be established and approved.

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MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.5.1	Housing, Electrical				9235541
					NEXT HIGHER ASSEMBLY 9235509
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of stator assembly hole		0.40%	3.1	Gage
102	True position of stator assembly hole with alternator assembly cavity		0.40%	3.1	Gage
103	Diameter of alternator assembly cavity		0.40%	3.1	Gage
104	True position of bearing plate cavity with center-line of dowel pin holes		0.40%	3.1	Gage
105	Location of dowel pin holes with side and end and true position of dowel pin holes with each other		0.40%	3.1	Gage
106	True position of bearing plate shoulder cavity with bearing plate cavity		0.40%	3.1	Gage
107	True position of alternator assembly cavity with bearing plate cavity		0.40%	3.1	Gage
Minor					
201	Evidence of poor workmanship		1.00%	3.5	Visual

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MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER 9235533
*4.4.5.2	Housing, Mechanical Drive				NEXT HIGHER ASSEMBLY 9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of gear assembly cavity		0.40%	3.1	Gage
102	Diameter of lower gear assembly bearing cavity		0.40%	3.1	Gage
103	Diameter of outer gear rack assembly cavity		0.40%	3.1	Gage
104	Diameter of inner gear rack assembly cavity		0.40%	3.1	Gage
105	True position of lower gear assembly bearing cavity with gear assembly cavity		0.40%	3.1	Gage
106	True position of outer gear rack assembly cavity with center-line of dowel pin holes		0.40%	3.1	Gage
107	True position of packing groove with outer gear rack assembly cavity		0.40%	3.1	Gage
108	Location of dowel pin holes with side and end and true position of dowel pin holes with each other		0.40%	3.1	Gage
109	Depth of smaller gear assembly cavity		0.40%	3.1	Gage
110	Diameter of smaller gear assembly cavity		0.40%	3.1	Gage
111	True position of smaller gear assembly cavity		0.40%	3.1	Gage
112	True position of larger gear assembly cavity		0.40%	3.1	Gage
113	Location of threaded cavity		0.40%	3.1	Gage
114	True position of inner gear rack assembly cavity with outer gear rack assembly cavity		0.40%	3.1	Gage

NOTES:

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MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 2 OF 2		DRAWING NUMBER
*4.4.5.2	Housing, Mechanical Drive				9235533
					NEXT HIGHER ASSEMBLY
					9235532
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Minor					
201	Diameter of packing groove		0.65%	3.1	Gage
202	Width of top segment		0.65%	3.1	Gage
203	Depth of upper gear assembly cavity		0.65%	3.1	Gage
204	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
*4.4.5.3	Case, Molding		9235505
			NEXT HIGHER ASSEMBLY 9235504

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Diameter of binding post holes		0.40%	3.1	Gage
102	Parallelism of sides of binding post hole cavity with center line		0.40%	3.1	Gage
103	Diameter of spring pin hole		0.40%	3.1	Gage
104	Location of binding post holes		0.40%	3.1	Gage
Minor					
201	Width of large cavity		0.65%	3.1	Gage
202	Length of large cavity		0.65%	3.1	Gage
203	Marking misleading or unidentifiable		0.65%	3.1	Visual
204	Serrations missing		0.65%	3.1	Visual
205	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-B-50451B (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
*4.4.5.4	Lever				9235508
					NEXT HIGHER ASSEMBLY
					9235504
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Symmetry of gear rack cutout with sides		0.40%	3.1	Gage
102	Symmetry of positioning slot with sides		0.40%	3.1	Gage
Minor					
201	Width of spring pin holes		0.65%	3.1	Gage
202	Total length		0.65%	3.1	Gage
203	Serrations missing		0.65%	3.1	Visual
204	Evidence of poor workmanship		1.00%	3.5	Visual

NOTES:

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4.5 Test methods and procedures. -

4.5.1 Capacity (output) test. - The blasting machine, with the specified load connected to its terminals, shall be actuated (see note) to produce the required output in accordance with 3.3.1. A photograph of the oscilloscope display of the waveform across the load shall be taken and observation made for conformance with the applicable requirement. Failure to produce the specified current for the minimum time shall be cause for rejection of the blasting machine.

NOTE - An actuation shall consist of rapidly depressing the handle (by hand or other equivalent mechanism) until an output is obtained. Up to a maximum of 5 successive handle depressions are permitted to obtain the required output.

4.5.2 Life cycle test. - The blasting machine shall be tested the required number of times in accordance with the test method specified in 4.5.1. During the test, a minimum of 6 photographs shall be taken of the output as displayed on an oscilloscope (output no. 1, 1000, 2000, 3000, 4000, 5000). Any evidence of wear, damage or failure to meet the requirements of 3.3.1 and 3.3.2 shall constitute failure of the test.

4.5.3 Humidity test. - The blasting machine shall be subjected to 20 hours exposure with relative humidity between 92 and 97 percent and at air temperature from plus 80 degrees F to plus 85 degrees F followed by 4 hours of 100 percent relative humidity with condensation at plus 75 degrees F. to plus 80 degrees F. At the conclusion of the humidity test the blasting machine shall be inspected for evidence of damage and tested for conformity with the requirements of 3.3.1.

4.5.4 Hardness test. - The applicable components shall be tested for hardness in accordance with FED-STD-151, Method 243.1.

4.5.5 Four-foot drop. - The blasting machine shall be dropped a total of three times, through each of its major axes, a vertical distance of four feet onto a solid concrete surface. One of said drops shall be with the terminal posts facing down. Following the test, the machine shall show no evidence of damage causing failure to conform to the requirements of 3.3.1.

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4.5.6 Insulation resistance. - The blasting machine shall be subjected to 500 volts DC applied for one minute between the shorted output terminals and the plunger. Following the test, the machine shall conform to the requirements of 3.3.1. Evidence of damage or dielectric failure shall be cause for rejection of the machines (See 3.3.5).

4.5.7 Leakage and operability test. - The blasting machine shall be completely submerged under 6 inches of water heated between 180 degrees F min and 190 degrees F max for a minimum of 5 minutes and a maximum of 10 minutes. Any blasting machine emitting a steady stream of bubbles shall be classed defective and removed from the lot. Blasting machines showing no evidence of leakage shall be operationally tested as specified in 4.5.1 for conformance with the requirements of 3.3.1.

4.5.8 Resistance test of stator assembly. - The output leads shall be connected to an ohmeter. Observation shall be made for compliance with the applicable drawing requirements.

4.5.9 Clutch torque test. - The clutch shall be assembled to a test fixture. The specified torque shall be applied and observation shall be made for compliance with the applicable drawing requirements.

4.5.10 Operating temperature test. -

4.5.10.1 Low temperature test. - The blasting machine shall be subjected to an air temperature of minus 35 degrees F to minus 40 degrees F until stabilized. While at this temperature, the blasting machine shall be tested as specified in 4.5.1 for conformance with the requirements of 3.3.1.

4.5.10.2 High temperature test. - The blasting machine shall be subjected to an air temperature of plus 120 degrees F to plus 125 degrees F until stabilized. While at this temperature, the blasting machine shall be tested as specified in 4.5.1 for conformance with the requirements of 3.3.1.

5. PREPARATION FOR DELIVERY

5.1 Packaging. -

5.1.1 Level A. - Fifty (50) blasting machines shall be packaged, one per cell, in a fiber box complete with half-

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slotted partitions assembled to form 25 cells in a 5 by 5 arrangement, 2 layers per box. A filler pad shall be placed on the inside bottom of the box, between layers and on top of the box contents. Additional pads shall be added as necessary to provide a tight package. The box shall conform to the requirements of PPP-B-636 for a Type CF, Class Domestic, Variety SW, Grade 200 box. The partitions and pads shall be constructed of the same test fiberboard as the box. Cell size shall be approximately 2 3/4 by 1 1/2 by 5 3/8 inches in depth inside dimensions. Box size shall be approximately 14 3/8 by 8 1/8 by 11 1/8 depth inside dimensions. Filler pads shall be equal to the inside length and width of the box less 1/16 inch. Boxes shall be closed by application of 3 inch wide, PPP-T-45 Type II, Class 1 or 2 tape over the full length of the top and bottom center seam of the box with an approximate 3 inch extension onto the end panels. All corners of the box shall be blunted and the box inserted into a barrier bag conforming to MIL-B-117 for a Type I, Style I, Class E bag. Bag size shall be approximately 21 by 25 inches inside dimensions. All excess air shall be exhausted from the bag and the opening closed by means of heatsealing. The closure seam shall meet the heatseal test contained in MIL-P-116.

5.1.2 Level C. - Blasting machines shall be packaged the same as Level A except the barrier bag shall be omitted.

5.2 Packing. -

5.2.1 Level A. - Two unit boxes (100 blasting machines) shall be packed in a Grade A wooden box conforming to PPP-B-621, for a Class 2, Style 2, 2 1/2 or 4 for a Type 3 load or MIL-B-46506 for a Type I Class 2, Grade A box. The interior surfaces of the box shall be lined with Filler Sheet Form, MIL-F-50449. One or more fillers shall be placed on top of the box contents to insure a tight pack. The box size shall be approximately 17 1/8 by 15 1/8 by 11 7/8 inches in depth, inside dimensions. Boxes shall be closed and strapped in accordance with instructions contained in the appendix of the box specification.

5.2.2 Level C. - The unit fiber box, without barrier bag, shall serve as the shipping container.

5.2.3 Air Shipment. - When air shipment is specified each box and each pallet load of boxes shall be tested by use of a Gauss meter to insure compliance with the magnetic shield-

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ing requirements for air shipment, as specified in MIL-S-4473. Boxes or pallet loads failing to meet the air suitability requirements shall be shielded in accordance with the requirements of MIL-S-4473 to meet air shipment requirements.

5.3 Marking. - Each unit box, barrier bag and exterior container shall be marked in accordance with MIL-STD-1169 requirements, except barrier bags shall be printed or stamped. In addition each exterior box shall have special magnetic caution labels applied in accordance with paragraphs 5.5.15.8.1 through 5.5.15.8.4 of MIL-STD-129. Labels shall be waterproofed in accordance with MIL-STD-129 requirements.

6. NOTES

6.1 Ordering data. - Procurement documents shall specify the following:

a. Title, number and date of this specification
b. Provisions for submission of first article samples.
c. Provisions of MIL-A-48078.

6.2 Submission of inspection equipment designs for approval (See MIL-A-48078). - Submit equipment designs as required, to Commander: Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Distribution of ammunition data cards. - Distribution of data cards shall include the following: Commander, Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover,New Jersey 07801.

6.4 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodian:
Army-PA

Preparing Activity:
Army-PA

Project Number: 1375-A-177

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

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☐ YES ☐ NO *(If "Yes", in what way?)*

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